

# Quantum Computing Text Lectures

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# 1 Mathematical Prerequisites: Complex Numbers

Before we begin exploring qubits, it is important to understand complex numbers. Complex numbers are central to how quantum states are defined and manipulated. Many of the fundamental ideas in quantum mechanics, such as superposition and measurement probabilities, rely on the properties of complex numbers.

## 1.1 Definition of a Complex Number

A complex number is a number that includes both a real part and an imaginary part. It is written in the form

$$z = x + iy \tag{1}$$

where:

- $x$  is the real part:  $x = \text{Re}(z)$
- $y$  is the imaginary part:  $y = \text{Im}(z)$
- $i$  is the imaginary unit:  $i = \sqrt{-1}$ , so  $i^2 = -1$

## 1.2 Complex Conjugate

Every complex number has an associated complex conjugate. If  $z = x + iy$ , then the complex conjugate of  $z$ , denoted  $z^*$ , is given by

$$z^* = x - iy. \tag{2}$$

The complex conjugate is obtained by changing the sign of the imaginary part.

## 1.3 Modulus

The modulus of a complex number represents its distance from the origin in the complex plane. The square of the modulus of  $z = x + iy$  is given by multiplying the number by its complex conjugate:

$$|z|^2 = zz^* = (x + iy)(x - iy) = x^2 + y^2. \tag{3}$$

Taking the square root gives the modulus:

$$|z| = \sqrt{x^2 + y^2}. \tag{4}$$

This formula is analogous to the Euclidean distance formula in two-dimensional space.

## 1.4 Example

Consider the complex number

$$z = 3 + 4i.$$

Its complex conjugate is

$$z^* = 3 - 4i.$$

The modulus squared is

$$|z|^2 = (3 + 4i)(3 - 4i) = 9 + 16 = 25.$$

Taking the square root, we find the modulus:

$$|z| = \sqrt{25} = 5.$$

## 2 Mathematical Prerequisites: Basics of Probability Theory

Probability theory forms the mathematical foundation for understanding measurement and uncertainty in quantum systems. In quantum mechanics, these ideas are fundamental, as the outcomes of quantum measurements are not deterministic but probabilistic.

We begin by describing the foundational elements of probability.

A *sample space*  $S$  is the collection of all possible outcomes of a random experiment. For instance, rolling a fair six-sided die produces

$$S = \{1, 2, 3, 4, 5, 6\},$$

where each number represents one possible outcome.

An *event* is a subset of the sample space. For example, the event “rolling an even number” corresponds to

$$A = \{2, 4, 6\}.$$

Each event is something we can assign a probability to.

The probability of an event can be interpreted as the relative frequency of its occurrence in repeated trials. Suppose that there are  $n$  total trials of an experiment, and the  $j^{\text{th}}$  event occurs  $n_j$  times. Then the probability that the  $j^{\text{th}}$  event occurs is:

$$p_j = \frac{n_j}{n}. \quad (5)$$

By definition, the sum of all probabilities over the full sample space must be 1:

$$\sum_{j=1}^{\infty} p_j = \sum_{j=1}^{\infty} \frac{n_j}{n} = \frac{1}{n} \sum_{j=1}^{\infty} n_j = \frac{n}{n} = 1. \quad (6)$$

A *random variable*  $X$  is a function that assigns a real number to each outcome in the sample space. For instance, if  $X$  maps the outcome of a die roll to the number shown, then

$$X(i) = i \quad (7)$$

for  $i \in \{1, 2, \dots, 6\}$ .

The *probability distribution* of  $X$  assigns a probability to each value it can take. For a fair die, the probability distribution is uniform:

$$P(X = x) = \frac{1}{6}, \quad \text{for } x = 1, 2, 3, 4, 5, 6.$$

To understand the long-term average outcome of  $X$ , we compute its *expected value*,

$$\mathbb{E}[X] = \sum_x x \cdot P(X = x). \quad (8)$$

This represents the mean value of  $X$  over many trials.

The *expectation of a function*  $f(X)$  tells us the average value of  $f(X)$  over many trials:

$$\mathbb{E}[f(X)] = \sum_x f(x) \cdot P(X = x).$$

If we are interested in how spread out the values of  $X$  are, we compute the *variance*:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \quad (9)$$

A larger variance means more unpredictability in the outcomes. The square root of the variance is the *standard deviation*, denoted by  $\sigma$ , which provides a measure of dispersion in the same units as  $X$ .

If event  $A$  has already occurred, and we want to know the probability that another event  $B$  also occurs, we compute the *conditional probability*:

$$P(B | A) = \frac{P(A \cap B)}{P(A)}, \quad \text{provided } P(A) > 0.$$

This tells us how the occurrence of  $A$  affects our belief about  $B$ . For example, if we know a die landed on an even number, the probability it was a 6 becomes  $\frac{1}{3}$ , not  $\frac{1}{6}$ .

Two events  $A$  and  $B$  are said to be *independent* if knowing that  $A$  has occurred does not affect the probability of  $B$ . That is, if:

$$P(B | A) = P(B),$$

then  $A$  and  $B$  are independent. Equivalently,  $P(A \cap B) = P(A)P(B)$ . Independence is a critical idea when analyzing systems where components evolve without influencing each other.

If we learn new information about an outcome, how do we update our beliefs? This is handled by *Bayes' theorem*, which tells us how to revise probabilities in light of new evidence. If event  $B$  has occurred, then the probability of  $A$  becomes:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}.$$

Bayes' theorem is widely used in statistical inference, machine learning, and quantum state estimation.

## Worked Example: Probability of Student Test Scores

Suppose a group of 6 students takes a test, and the scores they receive are:

$$\{60, 70, 70, 80, 90, 100\}$$

We define a discrete random variable  $X$  to represent the test score of a randomly chosen student. Let us compute the probability distribution, the expected score, variance, and examine some conditional and joint probabilities.

### Step 1: Sample Space and Frequency Table

| Score ( $x$ ) | Frequency ( $n_x$ ) | Probability $P(X = x) = \frac{n_x}{6}$ |
|---------------|---------------------|----------------------------------------|
| 60            | 1                   | $\frac{1}{6}$                          |
| 70            | 2                   | $\frac{1}{3}$                          |
| 80            | 1                   | $\frac{1}{6}$                          |
| 90            | 1                   | $\frac{1}{6}$                          |
| 100           | 1                   | $\frac{1}{6}$                          |

**Step 2: Expected Value (Mean)** The expected score is:

$$\begin{aligned}\mathbb{E}[X] &= \sum_x x \cdot P(X = x) = 60 \cdot \frac{1}{6} + 70 \cdot \frac{1}{3} + 80 \cdot \frac{1}{6} + 90 \cdot \frac{1}{6} + 100 \cdot \frac{1}{6} \\ &= \frac{60 + 140 + 80 + 90 + 100}{6} = \frac{470}{6} \approx 78.33\end{aligned}$$

**Step 3: Variance and Standard Deviation** We first compute  $\mathbb{E}[X^2]$ :

$$\begin{aligned}\mathbb{E}[X^2] &= 60^2 \cdot \frac{1}{6} + 70^2 \cdot \frac{1}{3} + 80^2 \cdot \frac{1}{6} + 90^2 \cdot \frac{1}{6} + 100^2 \cdot \frac{1}{6} \\ &= \frac{3600 + 2 \cdot 4900 + 6400 + 8100 + 10000}{6} = \frac{3600 + 9800 + 6400 + 8100 + 10000}{6} \\ &= \frac{37900}{6} \approx 6316.67\end{aligned}$$

Then,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 6316.67 - (78.33)^2 \approx 6316.67 - 6135.39 = 181.28$$

$$\sigma = \sqrt{181.28} \approx 13.46$$

**Step 4: Event Probabilities** Let  $A$  be the event “score  $\geq 80$ ”. Then:

$$P(A) = P(X = 80) + P(X = 90) + P(X = 100) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = 0.5$$

Let  $B$  be the event “score is even”, i.e.,  $B = \{60, 70, 80, 100\}$ . Then:

$$P(B) = P(X = 60) + P(X = 70) + P(X = 80) + P(X = 100) = \frac{1}{6} + \frac{1}{3} + \frac{1}{6} + \frac{1}{6} = \frac{5}{6}$$

**Step 5: Conditional Probability** Suppose we know the student scored at least 80 (event  $A$ ). What is the probability the student scored exactly 90 (event  $C = \{90\}$ )?

$$P(C \mid A) = \frac{P(C \cap A)}{P(A)} = \frac{P(90)}{P(A)} = \frac{1/6}{1/2} = \frac{1}{3}$$

**Step 6: Independence of Events** Are  $A$  (score  $\geq 80$ ) and  $B$  (score even) independent?

$$P(A \cap B) = P(X = 80) + P(X = 100) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6}$$

$$P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{5}{6} = \frac{5}{12}$$

Since  $\frac{2}{6} = \frac{1}{3} \neq \frac{5}{12}$ , the events  $A$  and  $B$  are not independent.

### 3 Quantum Mechanics Notation: Dirac Notation

Quantum mechanics uses a special notation, introduced by Paul Dirac, to represent quantum states and their relationships. This notation is known as Dirac notation, or bra-ket notation.

In Dirac notation, a quantum state is written as a ket:

$$|\psi\rangle.$$

This ket represents a vector in a complex vector space. The corresponding conjugate transpose of this vector is called a bra and is written as

$$\langle\psi|.$$

When a bra and a ket are combined, they form a bracket:

$$\langle\psi|\phi\rangle,$$

which represents the inner product between the two state vectors  $|\psi\rangle$  and  $|\phi\rangle$ . The result of this inner product is a complex number.

## 4 State Space Representation

An  $n$ -dimensional vector space  $V$  is a non-empty set with elements  $|u\rangle, |v\rangle$  called vectors. Vectors can be written in the following way:

$$|v\rangle = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \quad (10)$$

where  $v_1, v_2, \dots, v_n$  are the components or the vector  $|v\rangle$ .

A Hilbert space is a complete vector space with an inner product, where:

- Vector space means it allows vector addition and scalar multiplication.
- Inner product gives a way to define angles and lengths (like a dot product in 3D).
- Complete means all Cauchy sequences of vectors in the space converge to a vector in the space.

### 4.1 Vector operations

#### Vector addition

For two vectors  $|u\rangle$  and  $|v\rangle$ :

$$|u\rangle + |v\rangle = \begin{pmatrix} u_1 \\ u_1 \\ \vdots \\ u_n \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix} \quad (11)$$

#### Scalar Multiplication

For a scalar  $c$  and vector  $|v\rangle$ :

$$c|v\rangle = c \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{pmatrix} \quad (12)$$

#### Inner Product

The inner product between two states  $|u\rangle$  and  $|v\rangle$  is:

$$\langle u|v\rangle = \begin{pmatrix} u_1^* & u_2^* & \dots & u_n^* \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = u_1^*v_1 + u_2^*v_2 + \dots + u_n^*v_n \quad (13)$$

## 5 State Space Representation

In quantum mechanics, physical systems are described by vectors in a complex vector space known as a Hilbert space. This section introduces the mathematical structure of such spaces and basic operations on vectors.

### 5.1 Vector Space

A *vector space* is a mathematical structure where we can add vectors and multiply them by numbers (called scalars). These scalars usually come from the real numbers  $\mathbb{R}$  or complex numbers  $\mathbb{C}$ .

Vectors in such a space can be represented as column vectors:

$$|v\rangle = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \quad (14)$$

where  $v_1, v_2, \dots, v_n$  are complex numbers.

For a set of vectors to be called a vector space, it must satisfy the following rules:

- **Closure under addition and scalar multiplication:** If  $|u\rangle$  and  $|v\rangle$  are vectors in the space, then their sum  $|u\rangle + |v\rangle$  must also be in the space. Similarly, if  $c$  is a scalar and  $|v\rangle$  is a vector, then  $c|v\rangle$  must also be in the space.

*Example:* If  $|u\rangle = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$  and  $|v\rangle = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ , then  $|u\rangle + |v\rangle = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$  is also a vector. If  $c = 2$ , then  $2|v\rangle = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$  is also in the space.

- **Addition is associative and commutative:** The order in which vectors are added doesn't matter.  $(|u\rangle + |v\rangle) + |w\rangle = |u\rangle + (|v\rangle + |w\rangle)$  (associativity)  $|u\rangle + |v\rangle = |v\rangle + |u\rangle$  (commutativity)

*Example:* Let  $|u\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ ,  $|v\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ , and  $|w\rangle = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ . Then  $(|u\rangle + |v\rangle) + |w\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$  and  $|u\rangle + (|v\rangle + |w\rangle) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ .

- **Distributivity:** Scalar multiplication distributes over vector addition and field addition:

$$c(|u\rangle + |v\rangle) = c|u\rangle + c|v\rangle, \quad (a + b)|v\rangle = a|v\rangle + b|v\rangle$$

*Example:* Let  $a = 2$ ,  $b = 3$ , and  $|v\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ . Then  $(2 + 3)|v\rangle = 5 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$  and  $2|v\rangle + 3|v\rangle = \begin{pmatrix} 2 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$ .

- **Zero vector and additive inverse:** There is a special vector called the zero vector  $|0\rangle$  such that  $|v\rangle + |0\rangle = |v\rangle$  for any vector  $|v\rangle$ . Also, for every vector  $|v\rangle$ , there exists a vector  $-|v\rangle$  such that  $|v\rangle + (-|v\rangle) = |0\rangle$ .

*Example:* If  $|v\rangle = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$ , then  $|0\rangle = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$  and  $-|v\rangle = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$ . So,  $|v\rangle + (-|v\rangle) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ .

Note that the most common examples of vector spaces are  $\mathbb{R}^n$  and  $\mathbb{C}^n$ .  $\mathbb{R}^n$  is the space of all  $n$ -dimensional vectors with real number components. For example,  $\mathbb{R}^2$  contains vectors like  $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ .  $\mathbb{C}^n$  is the space of all  $n$ -dimensional vectors with complex number components.

## 5.2 Hilbert Space

A Hilbert space is a complex vector space equipped with an inner product that satisfies:

- Vector addition and scalar multiplication (linearity)
- An inner product to define angles and lengths
- Completeness: every Cauchy sequence converges within the space

The state space of a quantum system is modeled as a Hilbert space. For example, a single quantum state (qubit) resides in the space  $\mathbb{C}^2$ , while a system of  $n$  such states is described in the tensor product space  $\mathbb{C}^{2^n}$ .

### 5.2.1 Vector Addition

Given two vectors  $|u\rangle$  and  $|v\rangle$  in  $\mathbb{C}^n$ , their sum is defined as:

$$|u\rangle + |v\rangle = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix} \quad (15)$$

### 5.2.2 Scalar Multiplication

For a scalar  $c \in \mathbb{C}$  and a vector  $|v\rangle \in \mathbb{C}^n$ :

$$c|v\rangle = c \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{pmatrix} \quad (16)$$

### 5.2.3 Inner Product (Bra-Ket Notation)

The inner product of two vectors  $|u\rangle$  and  $|v\rangle$  is written as  $\langle u|v\rangle$  and is defined as:

$$\langle u|v\rangle = \begin{pmatrix} u_1^* & u_2^* & \dots & u_n^* \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = u_1^*v_1 + u_2^*v_2 + \dots + u_n^*v_n \quad (17)$$

Here,  $u_i^*$  denotes the complex conjugate of the  $i$ th component of  $|u\rangle$ . The inner product satisfies linearity in the second argument, conjugate symmetry, and positivity.

## 6 Basis Sets and Linear Independence

### 6.1 Span of a Set of Vectors

Given a set of vectors  $\{|v_1\rangle, |v_2\rangle, \dots, |v_n\rangle\}$ , any *linear combination* of these vectors is written as:

$$|\psi\rangle = \alpha_1 |v_1\rangle + \alpha_2 |v_2\rangle + \dots + \alpha_n |v_n\rangle = \sum_{i=1}^n \alpha_i |v_i\rangle$$

where  $\alpha_i$  are complex numbers.

We say that this set *spans* a vector space  $V$  if any vector  $|u\rangle$  in  $V$  can be written as a linear combination of the vectors in the set. In other words, the span is the collection of all vectors that can be formed from linear combinations of the given set.

For example, in the complex 3-dimensional vector space  $\mathbb{C}^3$ , any vector can be expressed as:

$$|u\rangle = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

So the set

$$|v_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |v_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |v_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

*spans*  $\mathbb{C}^3$ .

Similarly, in the qubit space  $\mathbb{C}^2$ , the standard basis states  $|0\rangle$  and  $|1\rangle$  span the space of all single-qubit states.

### 6.2 Linear Independence

Now that we understand span, we introduce the idea of linear independence. A set of vectors  $\{|v_1\rangle, |v_2\rangle, \dots, |v_n\rangle\}$  is said to be *linearly independent* if the only solution to the equation:

$$\alpha_1 |v_1\rangle + \alpha_2 |v_2\rangle + \dots + \alpha_n |v_n\rangle = 0$$

is  $\alpha_1 = \alpha_2 = \dots = \alpha_n = 0$ . That is, no vector in the set can be written as a linear combination of the others.

If, on the other hand, there exist coefficients—at least one non-zero—such that the linear combination equals the zero vector, then the vectors are *linearly dependent*. In this case, at least one vector can be expressed using the others.

### 6.3 Basis

A set of vectors forms a basis for a vector space  $V$  when two conditions are satisfied:

1. The vectors are linearly independent
2. The vectors span the entire space  $V$

The dimension of a vector space  $V$  equals the number of elements in any basis set for that space. For  $\mathbb{C}^2$ , the dimension is 2, which is why both our example basis sets contain exactly two vectors.

The set  $|0\rangle, |1\rangle$  forms a basis for  $\mathbb{C}^2$  because:

- **Spanning property:** Any vector in  $\mathbb{C}^2$  can be written as  $\alpha|0\rangle + \beta|1\rangle$ . When we have a basis, any vector in the space can be expressed as a unique linear combination of the basis vectors.
- **Linear independence:** The only solution to  $c_1|0\rangle + c_2|1\rangle = 0$  is  $c_1 = c_2 = 0$

## 7 Orthonormality

### 7.1 Vector Normalization

A vector  $|a\rangle$  is normalized when its norm equals unity:

$$\langle a|a\rangle = 1 \quad (18)$$

If a vector is not normalized, we can create a normalized version by dividing by its norm:

$$|\tilde{a}\rangle = \frac{|a\rangle}{\sqrt{\langle a|a\rangle}} \quad (19)$$

**Example:** Consider the vectors  $|u_1\rangle = |0\rangle + |1\rangle$  and  $|u_2\rangle = |0\rangle - |1\rangle$ . Their norms are:

$$||u_1\rangle| = ||u_2\rangle| = \sqrt{1^2 + 1^2} = \sqrt{2} \quad (20)$$

The normalized versions are:

$$|\tilde{u}_1\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |\tilde{u}_2\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad (21)$$

### 7.2 Orthonormal Sets

A set of vectors is orthonormal when:

1. Each vector is normalized (unit norm)
2. All pairs of distinct vectors are orthogonal (zero inner product)

### 7.3 The Standard Basis is Orthonormal

Let's verify that  $|0\rangle, |1\rangle$  is orthonormal:

**Normalization check:**

$$\langle 0|0\rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1^* \cdot 1 + 0^* \cdot 0 = 1 \quad (22)$$

$$\langle 1|1\rangle = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0^* \cdot 0 + 1^* \cdot 1 = 1 \quad (23)$$

**Orthogonality check:**

$$\langle 0|1\rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1^* \cdot 0 + 0^* \cdot 1 = 0 \quad (24)$$

$$\langle 1|0\rangle = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0^* \cdot 1 + 1^* \cdot 0 = 0 \quad (25)$$

Since both normalization and orthogonality conditions are satisfied,  $|0\rangle, |1\rangle$  is indeed orthonormal.

## 8 Operators in Quantum Mechanics

In quantum mechanics, operators play a central role in describing how quantum systems evolve and how measurements are performed. An operator is a mathematical rule that acts on a quantum state (represented by a ket) and transforms it into another state.

### 8.1 Definition of an Operator

An operator  $\hat{A}$  is a rule that maps a ket  $|\psi\rangle$  to another ket  $|\phi\rangle$ :

$$\hat{A}|\psi\rangle = |\phi\rangle \quad (26)$$

Operators can also act on bras:

$$\langle\mu|\hat{A} = \langle\nu| \quad (27)$$

In many important cases, operators transform vectors within the same Hilbert space.

### 8.2 Linearity of Operators

An operator  $\hat{A}$  is linear if, for any complex numbers  $\alpha, \beta$  and states  $|\psi_1\rangle, |\psi_2\rangle$ , we have:

$$\hat{A}(\alpha|\psi_1\rangle + \beta|\psi_2\rangle) = \alpha\hat{A}|\psi_1\rangle + \beta\hat{A}|\psi_2\rangle \quad (28)$$

More generally, if a vector is written as a linear combination of basis states:

$$|\psi\rangle = \sum_i \alpha_i |u_i\rangle \quad (29)$$

then the operator acts as:

$$\hat{A}|\psi\rangle = \sum_i \alpha_i \hat{A}|u_i\rangle \quad (30)$$

### 8.3 Special Operators

Some operators are used frequently in quantum mechanics:

- **Identity Operator  $\hat{I}$ :** Leaves the state unchanged.

$$\hat{I}|\psi\rangle = |\psi\rangle$$

- **Zero Operator  $\hat{0}$ :** Maps any state to the zero vector.

$$\hat{0}|\psi\rangle = 0$$

### 8.4 Observables as Operators

In quantum mechanics, physical quantities like position, momentum, energy, and spin are called *observables*. Each observable is associated with a unique operator. For example, the position observable corresponds to the position operator, the energy to the Hamiltonian operator, and so on.

## 9 Operators: Pauli Matrices

In the context of quantum computing, especially when working within the two-dimensional Hilbert space of a single quantum state, a particularly important set of operators arises: the *Pauli matrices*. These are three  $2 \times 2$  matrices denoted by  $X$ ,  $Y$ , and  $Z$ , which represent fundamental quantum operations on quantum states. They correspond to essential transformations such as bit flips and phase flips, and they form a basis for the space of Hermitian operators on a qubit (You will read about this more during Quantum gates).

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (\text{Identity}) \quad (31)$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (\text{Bit-flip or NOT gate}) \quad (32)$$

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (\text{Phase-flip with bit-flip}) \quad (33)$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (\text{Phase-flip}) \quad (34)$$

These operators act on the computational basis states  $|0\rangle$  and  $|1\rangle$  as follows:

$$\begin{aligned} I|0\rangle &= |0\rangle, & I|1\rangle &= |1\rangle \\ X|0\rangle &= |1\rangle, & X|1\rangle &= |0\rangle \\ Y|0\rangle &= i|1\rangle, & Y|1\rangle &= -i|0\rangle \\ Z|0\rangle &= |0\rangle, & Z|1\rangle &= -|1\rangle \end{aligned}$$

### Exercise: Properties of the Pauli Matrices

1. **Squaring:** Show that each Pauli matrix is its own inverse:

$$X^2 = Y^2 = Z^2 = I$$

where  $I$  is the  $2 \times 2$  identity matrix.

2. **Hermitian and Unitary Properties:**

- (a) Show that each Pauli matrix is Hermitian, i.e.,  $X^\dagger = X$ ,  $Y^\dagger = Y$ , and  $Z^\dagger = Z$ .
- (b) Show that each Pauli matrix is also unitary, i.e.,  $X^\dagger X = Y^\dagger Y = Z^\dagger Z = I$ .

3. **Commutation and Anticommutation:**

- (a) Verify the following multiplication relations:

$$\begin{aligned} XY &= iZ, & YZ &= iX, & ZX &= iY \\ YX &= -iZ, & ZY &= -iX, & XZ &= -iY \end{aligned}$$

(b) Use the above to derive the commutators:

$$[X, Y] = 2iZ, \quad [Y, Z] = 2iX, \quad [Z, X] = 2iY$$

(c) Also compute the anticommutators:

$$\{X, Y\} = XY + YX = 0, \quad \{Y, Z\} = 0, \quad \{Z, X\} = 0$$

**Hint:** Use matrix multiplication and properties of complex conjugate transpose (Hermitian adjoint).

## 10 Outer Products

The outer product of two states  $|\psi\rangle$  and  $|\phi\rangle$  written as  $|\psi\rangle\langle\phi|$ , is an operator. When applied to another state  $|\chi\rangle$ , we get:

$$|\psi\rangle\langle\phi||\chi\rangle = \langle\phi|\chi\rangle|\psi\rangle \quad (35)$$

This transforms the vector  $|\chi\rangle$  into a new vector that is proportional to  $|\psi\rangle$ , scaled by the inner product  $\langle\phi|\chi\rangle$ .

### 10.1 Example: Outer Product as an Operator

Let us define an operator using outer products:

$$\hat{O} = |0\rangle\langle 0| - |1\rangle\langle 1| \quad (36)$$

Acting on a general qubit state  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ :

$$\begin{aligned} \hat{O}|\psi\rangle &= (|0\rangle\langle 0| - |1\rangle\langle 1|)(\alpha|0\rangle + \beta|1\rangle) \\ &= \alpha|0\rangle\langle 0|0\rangle - \beta|1\rangle\langle 1|1\rangle \\ &= \alpha|0\rangle - \beta|1\rangle \end{aligned}$$

This operator  $\hat{O}$  has transformed  $|0\rangle \rightarrow |0\rangle$  and  $|1\rangle \rightarrow -|1\rangle$ . This is the action of the  $Z$  operator.

### 10.2 Closure Relation

The completeness relation (closure) states that the identity operator can be written in terms of an orthonormal basis  $\{|u_i\rangle\}$  as:

$$\hat{I} = \sum_i |u_i\rangle\langle u_i| \quad (37)$$

Using this, any state can be expanded as:

$$|\psi\rangle = \hat{I}|\psi\rangle = \sum_i |u_i\rangle\langle u_i|\psi\rangle \quad (38)$$

## 11 Matrix Representation of Operators

Operators in finite-dimensional vector spaces can be represented as matrices. In an  $n$ -dimensional space, operators are represented by  $n \times n$  matrices. If we know how the operator acts on a basis  $\{|u_i\rangle\}$ , we can write:

$$\hat{A} = \sum_{i,j} \langle u_i | \hat{A} | u_j \rangle |u_i\rangle \langle u_j| \quad (39)$$

The number  $\langle u_i | \hat{A} | u_j \rangle$  gives the matrix element in row  $i$  and column  $j$ .

### 11.1 Example: Matrix Representation of a Simple Operator

Suppose an operator  $\hat{M}$  acts on the basis states as:

$$\hat{M} |0\rangle = |0\rangle, \quad \hat{M} |1\rangle = -|1\rangle$$

Then its matrix representation in the  $\{|0\rangle, |1\rangle\}$  basis is:

$$\hat{M} = \begin{pmatrix} \langle 0 | \hat{M} | 0 \rangle & \langle 0 | \hat{M} | 1 \rangle \\ \langle 1 | \hat{M} | 0 \rangle & \langle 1 | \hat{M} | 1 \rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (40)$$

### 11.2 Outer Product as Matrix

Let us take two general qubit states:

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}, \quad |\phi\rangle = \begin{pmatrix} c \\ d \end{pmatrix}$$

Then the outer product  $|\psi\rangle \langle \phi|$  is a matrix:

$$|\psi\rangle \langle \phi| = \begin{pmatrix} a \\ b \end{pmatrix} (c^* \quad d^*) = \begin{pmatrix} ac^* & ad^* \\ bc^* & bd^* \end{pmatrix} \quad (41)$$

This matrix acts as an operator in the 2D Hilbert space  $\mathbb{C}^2$ .

## 12 Eigenvalues and Eigenvectors

Let  $\hat{A}$  be an operator. A vector  $|\psi\rangle$  is an eigenvector of  $\hat{A}$  if it satisfies:

$$\hat{A}|\psi\rangle = \lambda|\psi\rangle \quad (42)$$

where  $\lambda$  is a scalar called the eigenvalue corresponding to  $|\psi\rangle$ .

### Finding Eigenvalues

To find the eigenvalues of an operator represented by a matrix  $A$ , solve the characteristic equation:

$$\det(A - \lambda I) = 0 \quad (43)$$

For a  $2 \times 2$  matrix:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

we have:

$$\det(A - \lambda I) = (a - \lambda)(d - \lambda) - bc = 0$$

### Finding Eigenvectors

Once the eigenvalues are found, substitute each eigenvalue  $\lambda$  into the equation:

$$(A - \lambda I)|\psi\rangle = 0$$

to solve for the components of the corresponding eigenvector  $|\psi\rangle$ . Then apply the normalization condition:

$$\langle\psi|\psi\rangle = 1$$

### Exercise

Find the eigenvalues and eigenvectors of the three Pauli matrices:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

### Solution:

1. Eigenvector-eigenvalue equation for  $X$ ,

$$X|\psi\rangle = \lambda|\psi\rangle$$

- Solve the characteristic equation:

$$\begin{aligned}(X - \lambda I) &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} \\ \det(X - \lambda I) &= \det \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 - 1 \\ \Rightarrow \lambda &= \pm 1\end{aligned}$$

Now solve  $(X - \lambda I)\vec{v} = 0$

- For  $\lambda = 1$ :

$$\begin{aligned}\begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} &= \vec{0} \\ \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} &= \vec{0} \\ \Rightarrow -a + b &= 0 \\ \Rightarrow a &= b\end{aligned}$$

One possible set of solutions is  $a = 1, b = 1$ . Applying normalization condition  $|a|^2 + |b|^2 = 1$  we get:

$$\begin{aligned}a &= \frac{a}{\sqrt{|a|^2 + |b|^2}} = \frac{1}{\sqrt{2}} \\ b &= \frac{b}{\sqrt{|a|^2 + |b|^2}} = \frac{1}{\sqrt{2}}\end{aligned}$$

$$\Rightarrow \text{Eigenvector: } |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (44)$$

- For  $\lambda = -1$ :

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \vec{0} \Rightarrow a + b = 0 \Rightarrow a = -b$$

Choose  $a = \frac{1}{\sqrt{2}}, b = -\frac{1}{\sqrt{2}}$ :

$$\Rightarrow \text{Eigenvector: } |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

---

## 2. Eigenvalues and Eigenvectors of $Y$

$$Y|\psi\rangle = \lambda|\psi\rangle$$

- Solve the characteristic equation:

$$(Y - \lambda I) = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} -\lambda & -i \\ i & -\lambda \end{pmatrix}$$

$$\det(Y - \lambda I) = \det \begin{pmatrix} -\lambda & -i \\ i & -\lambda \end{pmatrix} = \lambda^2 - 1$$

$$\Rightarrow \lambda = \pm 1$$

Now solve  $(Y - \lambda I)\vec{v} = 0$

- For  $\lambda = 1$ :

$$\begin{pmatrix} -1 & -i \\ i & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \vec{0}$$

$$\Rightarrow -a - ib = 0$$

$$\Rightarrow a = -ib$$

One possible set of solutions is  $a = -i, b = 1$ . Applying normalization condition  $|a|^2 + |b|^2 = 1$ , we get:

$$a = \frac{-i}{\sqrt{2}}, \quad b = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \text{Eigenvector: } |y_+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ 1 \end{pmatrix} \quad (45)$$

- For  $\lambda = -1$ :

$$\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \vec{0}$$

$$\Rightarrow a - ib = 0$$

$$\Rightarrow a = ib$$

One possible set of solutions is  $a = i, b = 1$ . Applying normalization:

$$a = \frac{i}{\sqrt{2}}, \quad b = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \text{Eigenvector: } |y_-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \end{pmatrix} \quad (46)$$

### 3. Eigenvalues and Eigenvectors of $Z$

$$Z|\psi\rangle = \lambda|\psi\rangle$$

- Solve the characteristic equation:

$$\begin{aligned}(Z - \lambda I) &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 1 - \lambda & 0 \\ 0 & -1 - \lambda \end{pmatrix} \\ \det(Z - \lambda I) &= (1 - \lambda)(-1 - \lambda) = \lambda^2 - 1 \\ &\Rightarrow \lambda = \pm 1\end{aligned}$$

Now solve  $(Z - \lambda I)\vec{v} = 0$

- For  $\lambda = 1$ :

$$\begin{aligned}\begin{pmatrix} 0 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} &= \vec{0} \\ &\Rightarrow b = 0\end{aligned}$$

One possible set of solutions is  $a = 1, b = 0$ . Normalize:

$$a = 1, \quad b = 0$$

$$\Rightarrow \text{Eigenvector: } |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{47}$$

- For  $\lambda = -1$ :

$$\begin{aligned}\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} &= \vec{0} \\ &\Rightarrow a = 0\end{aligned}$$

One possible set of solutions is  $a = 0, b = 1$ . Normalize:

$$a = 0, \quad b = 1$$

$$\Rightarrow \text{Eigenvector: } |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{48}$$

## 13 Hermitian and unitary operators

### 13.1 Hermitian Operators

An operator  $\hat{A}$  is said to be Hermitian (or self-adjoint) if it is equal to its own Hermitian adjoint:

$$\hat{A} = \hat{A}^\dagger$$

The Hermitian adjoint (also called the *conjugate transpose*) of an operator is defined by the condition:

$$\langle a | \hat{A}^\dagger | b \rangle = \left( \langle b | \hat{A} | a \rangle \right)^*$$

#### Properties of Hermitian Operators

- All eigenvalues of Hermitian operators are real.
- Eigenvectors corresponding to distinct eigenvalues are orthogonal.
- Hermitian operators represent observables in quantum mechanics, such as position, momentum, and energy.

#### Matrix Representation

For a  $2 \times 2$  matrix:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the Hermitian adjoint is:

$$A^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}$$

Thus,  $A$  is Hermitian if  $A = A^\dagger$ , that is,

$$\begin{aligned} a &= a^*, & d &= d^* & (\text{diagonal elements are real}) \\ c &= b^* & (\text{off-diagonal elements are complex conjugates}) \end{aligned}$$

.

### Note: Rules for Computing the Hermitian Adjoint

To compute the Hermitian adjoint (also called the dagger or conjugate transpose) of any quantum expression:

- Take the complex conjugate of all scalar coefficients.
- Replace all kets  $|\psi\rangle$  with bras  $\langle\psi|$ .
- Replace all bras  $\langle\psi|$  with kets  $|\psi\rangle$ .
- Replace each operator  $\hat{A}$  with its adjoint  $\hat{A}^\dagger$ .
- If the expression contains a product of operators or states, reverse the order in the adjoint.
- The Hermitian adjoint of a sum of operators is equal to the sum of their respective adjoints:

$$(\hat{A} + \hat{B} + \hat{C})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger + \hat{C}^\dagger$$

This follows from the linearity of the adjoint operation.

These rules are encapsulated in the following identities:

$$\begin{aligned}(\alpha\hat{A})^\dagger &= \alpha^* \hat{A}^\dagger \\ (|\psi\rangle)^\dagger &= \langle\psi| \\ (\langle\psi|)^\dagger &= |\psi\rangle \\ (\hat{A}\hat{B})^\dagger &= \hat{B}^\dagger \hat{A}^\dagger \\ (\hat{A}|\psi\rangle)^\dagger &= \langle\psi| \hat{A}^\dagger \\ (\hat{A}\hat{B}|\psi\rangle)^\dagger &= \langle\psi| \hat{B}^\dagger \hat{A}^\dagger\end{aligned}$$

## 13.2 Unitary Operators

An operator  $\hat{U}$  is said to be unitary if its adjoint is equal to its inverse:

$$\hat{U}^\dagger = \hat{U}^{-1}$$

This implies the defining condition:

$$\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \hat{I}$$

### Properties of Unitary Operators

- Unitary operators preserve the norm of quantum states.
- They describe time evolution and quantum gates in quantum computing.

- All eigenvalues of unitary operators lie on the complex unit circle, i.e., if  $\lambda$  is an eigenvalue, then  $|\lambda| = 1$ .

### 13.3 Normal Operators

An operator

$$\hat{A}$$

is said to be **normal** if it commutes with its adjoint:

$$\hat{A}^\dagger \hat{A} = \hat{A} \hat{A}^\dagger$$

All Hermitian and Unitary operators are special cases of normal operators.

## 14 Tensor Product

The tensor product is a mathematical operation used to combine two or more quantum systems into a single composite system. It is fundamental to the structure of quantum mechanics, especially when dealing with systems composed of multiple particles or subsystems.

### 14.1 Tensor Product of Quantum States

Let  $|\phi\rangle$  and  $|\psi\rangle$  be two individual quantum states. The combined system is described by the tensor product:

$$|\phi\rangle \otimes |\psi\rangle$$

This product produces a new state in a higher-dimensional space, representing the joint configuration of the two systems.

**Example:**

$$|\phi\rangle = |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\psi\rangle = |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Then their tensor product is:

$$|0\rangle \otimes |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 0 \\ 1 \cdot 1 \\ 0 \cdot 0 \\ 0 \cdot 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |01\rangle$$

This new state,  $|01\rangle$ , belongs to a four-dimensional space that represents all possible combinations of two two-level quantum systems.

### 14.2 Tensor Product of Operators

Just as quantum states are combined using the tensor product, operators acting on these states are also combined in the same way.

Let  $A$  and  $B$  be  $2 \times 2$  matrices (linear operators). Then:

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{pmatrix}$$

**Example:** Let

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Then:

$$A \otimes B = \begin{pmatrix} 0 \cdot B & 1 \cdot B \\ 1 \cdot B & 0 \cdot B \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

This operator acts on a composite state in a four-dimensional space, transforming both parts of the system.

### 14.3 Applying Operators to Composite Quantum States

Suppose we have a quantum state:

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

And we want to apply the following matrix to the first subsystem:

$$M = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

To apply it to the full system, we use the tensor product:

$$M \otimes I = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}$$

Now we apply this to the composite state:

$$(M \otimes I) |00\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$$

### 14.4 Properties of the Tensor Product

Let  $A, B, C, D$  be matrices and  $|v\rangle, |w\rangle$  be state vectors. Then the tensor product satisfies:

- Associativity:  $(A \otimes B) \otimes C = A \otimes (B \otimes C)$
- Distributivity:  $A \otimes (B + C) = A \otimes B + A \otimes C$
- Compatibility with scalar multiplication:  $(a|v\rangle) \otimes |w\rangle = a(|v\rangle \otimes |w\rangle)$
- Adjoint property:  $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$

The tensor product is central to building larger quantum systems and understanding interactions between different parts of a system. It also plays a critical role in defining concepts like entanglement and separability, which are fundamental to quantum information theory.